Project Proposal: Physical Consistency of Video Generation Models

A Two-Stage Benchmark on Open-Source Text-to-Video Models

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1. Problem Statement

State-of-the-art text-to-video (T2V) models (Sora, Runway, Kling) produce visually stunning output, yet recent benchmarks — Physics-IQ, VideoPhy-2, PhyGenBench, T2VPhysBench, Principia — show that *visual plausibility does not imply physical correctness*: SOTA generators violate gravity, momentum conservation, and contact mechanics while scoring high on visual quality. This breaks the downstream uses for which video generators are framed as "world models" (robot simulation, embodied-AI data augmentation). This project asks: **how physically consistent are open-source T2V models, measured fully locally and reproducibly at zero API cost? Part A (measurable physics):** generate simple phenomena with exactly known physics (free fall, projectile, pendulum, horizontal motion with friction), extract the object trajectory frame-by-frame, and fit physical parameters (g, pendulum period T, deceleration). **Part B (complex scenes):** generate collisions, fluids, support, multi-object interactions, then count physical violations (floating, interpenetration, wrong collision outcomes) with a vision-language model (VLM) judge under a fixed protocol.

2. Data

All data is generated locally on one RTX 5090 (32 GB), not downloaded.

- **Models**: Wan2.1-T2V-1.3B (Apache-2.0, 29 GB, 480p@16 fps, 81 frames) and CogVideoX-5B, open on Hugging Face; fallback Wan2.2-TI2V-5B.
- **Prompt sets**: Part A: 4 prompts, each a single red ball (stated 10 cm diameter; 1 m pendulum string — the pixel-to-meter calibration) on a plain white background. Part B: 8 prompts, realistic fixed-camera scenes, no text. Each prompt × 3 seeds × 2 models = 72 videos.
- **Ground-truth anchors**: (i) synthetic animations rendered with exact kinematics (g=9.8) to validate the trajectory detector before it touches generated video; (ii) 4 calibration prompts with known violations (person floating mid-air; glass suspended unsupported) that the VLM judge must flag, or the judging protocol is declared invalid.
- **Obstacles**: per-video generation cost (minutes), detector failures on artifact-heavy frames, judge subjectivity. Mitigations: 72-video scale, explicit detection-failure signaling instead of garbage fits, calibration-gated judging.

3. Methods

A Python pipeline ("physcheck"): PyTorch/diffusers for generation, OpenCV/SciPy for detection, pluggable VLM for judging.

(a) Generation. Standard diffusers pipelines (WanPipeline, CogVideoXPipeline), fixed seeds, metadata sidecars (fps, frames, commit hash) for traceability.

(b) Part A detector. Median-frame background subtraction isolates the ball; the largest connected component gives centroid and pixel diameter per frame. The centroid sequence is the trajectory; the diameter plus the known object size gives the pixel-to-meter scale. Least-squares fits: $y(t) = y_0 + v_0 t + \frac{1}{2} g t^2$ (fall), a parabola (projectile), $x(t) = A\cos(\sqrt{g/L}\, t + \phi)$ (pendulum), $x(t) = v_0 t - \frac{1}{2} a t^2$ (friction). R^2 and pixel-RMS gates reject garbage fits. *Validation:* the detector must recover $g \approx 9.8$ and the correct period from the synthetic animations before analyzing generated video.

(c) Part B judge. Each video becomes an 8-frame annotated grid; a VLM judge (Claude by default, GPT-4o/Gemini swappable) answers a fixed template and outputs JSON: per violation — frame index, category (gravity/support/collision/fluid/rigid-body), description, confidence. Judging is blind to model identity, in isolated calls.

Relation to existing work: we reuse open-weight models and the VLM-judge methodology of VideoPhy/PhyGenBench unmodified, but those benchmarks target commercial APIs and cannot be reproduced locally. Our contribution: the objective Part A (programmatic physics fitting, done by no prior video benchmark) plus a calibration-gated local judging protocol.

4. Evaluation and Expected Results

Quantitative. Part A: fitted g (m/s^2) and T (s) with standard errors from the fit covariance, relative error vs. truth, track/fit success rates. Part B: violation rate, mean violations per video, per-category counts; leaderboard across models; seed spread as uncertainty. Pipeline-trust evidence: the detector-validation table (true vs. recovered g, T on synthetic animations) and the calibration-gate result.

Qualitative. Figures: (i) extracted trajectories overlaid with fitted parabola/sine curves; (ii) Part B frame grids with judge annotations; (iii) leaderboard tables and violation heatmaps.

Hypotheses. (H1) Open T2V models do *not* encode quantitative physics: fitted g deviates from 9.8 m/s^2 by tens of percent; periods contradict $2\pi\sqrt{L/g}$ — diffusion imitates appearance, not dynamics. (H2) Part B violation rates high for collision/support, lower for fluids, mirroring commercial-model findings. (H3) The detector recovers synthetic ground truth within 10%. (H4) The VLM judge passes the calibration gate, supporting its use as a reproducible metric. Versus commercial models on Physics-IQ/VideoPhy, open models should be comparable or worse in violation rate — but, unlike those baselines, measurable in physical units for the first time.

Work products: "physcheck" benchmark code (MIT), prompt sets, leaderboard, paper-draft report; all experiments local, zero API cost.